

Communications and Networks (Spring 2022)

Problem Set 5

Issued: May 13 Due: May 22

Readings: Haykin - Communication System 5th ed. 10.11

1. In a single-parity-check code, a single parity bit b_1 is appended to a block of k information bits, $m_1, m_2, \dots, m_k$. The parity bit is chosen so that the codeword satisfies the even parity rule:

$$m_1 + m_2 + \dots + m_k + b_1 = 0, \quad \text{mod } 2$$

- a) For $k = 3$, construct the 2^k valid codewords in the code defined by this rule.
- b) Show that the minimum Hamming distance among valid codewords is $d_{\min} = 2$.
2. Consider a linear block code with total length $n = 13$ bits. If the code tolerates up to 3 bit errors, at least how many parity check bits is needed?
- (Hint: You need not construct the specific code)
3. Consider a (7,4) Hamming code with generator matrix

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

- a) Determine the parity check matrix $\mathbf{H}$.
- b) Decode the message if the received codeword is 0101011.
4. Consider a (7,3) linear block code with generator matrix

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- a) Construct the generator matrix $\bar{\mathbf{G}}$ of an equivalent systematic code and brief on why they are equivalent. (Hint: Conduct row transformations on $\mathbf{G}$)
- b) Determine the minimum Hamming distance $d_{\min}$. What is the error correction capacity t and the error detection capacity e ?
- c) Can you decode the message if the received codeword is $r_1 = 0100010$ or $r_2 = 1111001$?